

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 9

Status: active BA2 execution plan after BA2-T1 proposition-structure lower-bound derivation.

Supersedes: Revision 8 only for forward execution state; R1-R8 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0 responsibility/non-goals: CLOSED.
- BA1 minimal BAE identity ontology: CLOSED.
- BAREferent and BAProposition: ACCEPTED first-class semantic identity families.
- ThreatForge is a case study/tooling instrument, not DDTA semantic authority.

2. Closed BA1 identity boundary

BA1 accepts exactly:

BAREferent

BAProposition

No Action, Relation, Participation, Role, Behavior, Information, Store, Contract, Boundary or State identity family may be added by BA2 unless a concrete counterexample satisfies the BA1 reopen criteria.

3. BA2 objective

BA2 defines the smallest stable methodology-neutral proposition/relation/action semantics over the BA1 identities while preserving:

semantic identity family

!=

semantic operator / participation role / modifier

!=

lexical label / authoring synonym

BA2 must support mechanical selection/projection without importing threat-method categories or requiring consumers to reinterpret raw prose.

4. BA2-T1 - proposition shape and participation-role lower bound

Status: COMPLETED / PROVISIONAL CANDIDATE.

BA2-T1 tested universal binary SPO, positional n-ary tuples and fixed domain/method slots against facial-access and order-fulfillment evidence.

Result

Pure binary SPO as universal core	REJECTED
Binary propositions as special cases	SUPPORTED
Anonymous positional participation	REJECTED
Role-bound n-ary participation	FORCED CANDIDATE
Explicit semantic operator	FORCED CANDIDATE
Scoped semantic modifier capability	REQUIRED / ENCODING OPEN
Third BA1 identity family	NOT FORCED

Active lower-bound candidate:

```
BAProposition
|- semanticOperator      1
|- participation         1..*
|   |- role              1
|   `-- term             1
`-- scopedModifier      0..*
```

At least one participant term is a BAReferent. Typed local values may appear when independent semantic identity is not required.

Referent classification remains separate from contextual proposition role. BA2-T1 supports explicit method-neutral classification capability but does not freeze its vocabulary or cardinality.

5. BA2-T2 - operator, role and modifier vocabulary pressure test

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

BA2-T2 must attempt to derive the smallest reusable methodology-neutral vocabulary needed to instantiate the T1 structure.

Pressure areas:

1. operator families needed by both corpora without copying document predicate labels mechanically;
2. role semantics for actor/executor, source/destination, information/object, request/result, correlation/context, provider/consumer/responsibility and affected state;
3. whether roles are globally canonical or operator-scoped;
4. operator-role compatibility/cardinality constraints;
5. representation boundary between role-bound participants and scoped modifiers;
6. condition, state, failure, applicability, realization, atomicity/concurrency and negation/polarity pressure;
7. reusable referent classification needed for mechanical human/method projections;
8. bounded method-consumer mapping without STRIDE/DFD leakage.

Exit condition

Produce a falsifiable vocabulary candidate and explicit rejected/deferred alternatives. Do not close BA2 if important operator/role/modifier ambiguity remains.

6. Later BA2 closure sequence - provisional only

If BA2-T2 yields a stable vocabulary candidate, later microsteps should pressure-test:

- cross-corpus regression;
- lexical canonical label versus synonym separation;
- role/cardinality consistency;
- whether any relation/action semantic identity needs a registry-level stable key;
- BA2 closure and reopen criteria.

The exact number of later microsteps is evidence-driven, not precommitted.

7. BA3 - derivation, provenance and authority

Only after BA2 closure, BA3 must close:

- document-to-BAReferent/BAProposition derivation rules;
- source locators;
- grounded, derived and diagnostic/unresolved materialization;
- identity/equivalence/lifecycle rules;

- acceptance/rejection/stale semantics;
- source-to-analysis and analysis-to-source traceability.

Governed documentation remains project authority.

8. BA4 - projections

Define derived human and method projections from the same accepted BA semantics. No view becomes source authority.

9. BA5 - vocabulary and optional assistance

Define lexical controlled vocabulary, authoring synonyms and optional assistance only after semantic operator/role contracts are closed. NLP/LLM proposals remain candidate-producing support and cannot establish project semantics silently.

10. BA6 - Base Analysis closure and regression

Regress the implemented Base Analysis design against closed corpora and at least one structurally different holdout. BA6 closes the complete Base Analysis design.

11. Analysis envelope remains downstream

Only after Base Analysis closure:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration;
- formal methodology overlays and STRIDE/STRIDE-AI evaluations.

The earlier STRIDE transfer probe remains only a bounded consumer probe.

12. Next authorized microstep

After the BA2-T1 package is reviewed, committed and remotely verified, execute only:

BA2-T2 - semantic operator, participation-role and scoped-modifier vocabulary pressure test.

Do not start BA3, formal STRIDE overlay design, Common Finding schema or implementation work.